

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 5

Use any method to determine if the series converges or diverges. Give reasons for your answer.

$$\sum_{n=1}^{\infty} \frac{n^{\sqrt{2}}}{5^n}$$

The nth-Term Test states that $\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero. Apply the nth-Term Test to the given series.

Notice that if the limit of $\lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}}}{5^n}$ is evaluated as n approaches ∞ , the indeterminate form $\frac{\infty}{\infty}$ results.

Apply l'Hôpital's rule to $\lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}}}{5^n}$.

$$\lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}}}{5^n} = \lim_{n \rightarrow \infty} \frac{\sqrt{2} n^{\sqrt{2}-1}}{(\ln 5) \cdot 5^n}$$

The limit $\lim_{n \rightarrow \infty} \frac{\sqrt{2} n^{\sqrt{2}-1}}{(\ln 5) \cdot 5^n} = \frac{\infty}{\infty}$ is in the indeterminate form. Apply l'Hôpital's rule again to $\lim_{n \rightarrow \infty} \frac{\sqrt{2} n^{\sqrt{2}-1}}{(\ln 5) \cdot 5^n}$.

$$\lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}}}{5^n} = \frac{\sqrt{2}}{\ln 5} \cdot \lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}-1}}{5^n}$$

Separate out the constants.

$$= \frac{\sqrt{2}}{\ln 5} \cdot \lim_{n \rightarrow \infty} \frac{(\sqrt{2}-1) n^{\sqrt{2}-2}}{\ln(5) \cdot 5^n}$$

Apply l'Hôpital's rule.

Simplify.

$$\lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}}}{5^n} = \frac{\sqrt{2}}{\ln 5} \cdot \lim_{n \rightarrow \infty} \frac{(\sqrt{2}-1) n^{\sqrt{2}-2}}{\ln(5) \cdot 5^n}$$

$$= \frac{\sqrt{2}}{\ln 5} \cdot \frac{\sqrt{2}-1}{\ln 5} \cdot \lim_{n \rightarrow \infty} \frac{n^{\sqrt{2}-2}}{5^n}$$

Separate out the constants.

$$= \frac{\sqrt{2}}{\ln 5} \cdot \frac{\sqrt{2}-1}{\ln 5} \cdot \lim_{n \rightarrow \infty} \left(\frac{1}{5^n} \right) \cdot \lim_{n \rightarrow \infty} (n^{\sqrt{2}-2})$$

Apply the product rule.

$$= \frac{\sqrt{2}}{\ln 5} \cdot \frac{\sqrt{2}-1}{\ln 5} \cdot 0 \cdot 0$$

Apply limits.

$$= 0$$

Simplify.

Thus, the nth-Term Test is inconclusive since the limit of the nth term is 0. Now apply the Ratio Test to the given series.

Let $\sum a_n$ be any series and suppose that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho$. The Ratio Test states that the series converges absolutely if $\rho < 1$ and diverges if $\rho > 1$ or ρ is infinite. The test is inconclusive if $\rho = 1$.

Substitute in the appropriate symbolic terms of the series into the equation for the Ratio Test.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^{\sqrt{2}}}{5^{n+1}}}{\frac{n^{\sqrt{2}}}{5^n}} \right|$$

Simplify the expression.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left| \frac{\frac{(n+1)^{\sqrt{2}}}{5^{n+1}}}{\frac{n^{\sqrt{2}}}{5^n}} \right| &= \lim_{n \rightarrow \infty} \left| \frac{(n+1)^{\sqrt{2}}}{5^{n+1}} \cdot \frac{5^n}{n^{\sqrt{2}}} \right| \\
 &= \lim_{n \rightarrow \infty} \left| \frac{5^n}{5^{n+1}} \cdot \frac{(n+1)^{\sqrt{2}}}{n^{\sqrt{2}}} \right| \\
 &= \lim_{n \rightarrow \infty} \left| \frac{1}{5} \left(1 + \frac{1}{n} \right)^{\sqrt{2}} \right|
 \end{aligned}$$

Now find $\lim_{n \rightarrow \infty} \frac{1}{n}$.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Substitute $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ back into the expression $\lim_{n \rightarrow \infty} \left| \frac{1}{5} \left(1 + \frac{1}{n} \right)^{\sqrt{2}} \right|$ and simplify.

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \left| \frac{1}{5} \left(1 + \frac{1}{n} \right)^{\sqrt{2}} \right| &= \frac{1}{5} (1 + 0)^{\sqrt{2}} \\
 &= \frac{1}{5}
 \end{aligned}$$

Thus, $\rho = \frac{1}{5}$. Because $\rho < 1$, the series converges absolutely by the Ratio Test. Since the series converges absolutely, by the Absolute Convergence Test it converges.